MODELLING CONCEPTS

Definitions

- **OOAD:-** is a software engineering approach that models a system as a group of interacting objects.
- It is used to model software systems as collections of co-operating objects, treating individual objects as instances of class within class hierarchy.

Definitions contd…

- **OOA:-** is a method of analysis that examines the requirements of a system from the classes and objects found in vocabulary of problem domain.
- OOA emphasizes building of real-world models using an object-oriented view.
- Also, OOA applies object modeling techniques to analyze function requirements of a system.
- OOA focuses on **WHAT** system does.

Definitions contd…

- **OOD:-** is a method of design encompassing the process of object-oriented decomposition and a notation for describing both logical & physical as well as static & dynamic models of system under design.
- OOD elaborates the analysis models to produce implementation specifications.
- OOD focuses on **HOW** system does it.
- **OML:-** is a standardized set of symbols and ways of arranging them to model an object oriented S/W design.

MODELING

- Model:- is pattern, plan ,representation , or description designed to show main object or workings of an object, system or concept.

IMPORTANCE

- Modeling is a proven and well–accepted engineering technique.

  For e.g. we build architectural models for building houses and high- rises...

  And mathematical models for analyzing effects of winds or earthquakes on buildings.

  Modeling is also used in manufacturing automobiles and electrical devices from microprocessors to telephone switching systems.

- ***It means modeling is nothing but a simplification of reality.***

Contd..

- A model provides blue-prints of a system. i.e. it provides a **detailed plan** of action.
- A model may be structural or behavioral.
- The basic reason to model is:- **to understand the system we are developing in better way.**

OBJECTIVES

- Helps us to visualize a system as it is or as we want it to be.
- Permits us to specify the structure or behavior of a system.
- Gives us a template that guides us in constructing a system.
- It documents the decisions that we have made.

PRINCIPLES

- The choice of what models to create has a profound influence on how problem is attacked and how a solution is shaped.
- Every model may be expressed at different levels of precision.
- The best models are connected to reality.
- No single model is sufficient. Every nontrivial system is best approached through a small set of nearly independent models.

Similarly, to understand the architecture of a system, we need several complementary and interlocking views:-

Use Case view

Design View

Process View

Implementation View

Deployment View

Together these views represent the blueprints of a software.

Object oriented methodology

- It is a set of methods, models, and rules for developing systems
- Many methodologies are available to choose from
- Difference lies in documentation of information & modeling notations and languages.

OBJECT ORIENTED METHODOLOGIES

- **Rambaugh's Object Modeling Technique** (well suited for describing the object model or the static structure of the system).

- **The Booch Methodology** (produces detailed object oriented design models).

- **The Jacobson Methodology** (well suited for producing use-case driven analysis models).

All the three methodologies are the origins of a UML.

UML FUNDAMENTALS

Unified Modelling Language

- Unified because it …– Combines main preceding OO methods (Booch by Grady Booch, OMT by Jim Rumbaugh and OOSE by Ivar Jacobson)
- Modelling because it is …– Primarily used for visually modelling systems. Many system views are supported by appropriate models
- Language because …– It offers a syntax through which you express modelled knowledge

What is UML?

- A language for capturing and expressing knowledge
- A tool for system discovery and development
- A tool for visual development modelling
- A set of well-founded guidelines
- A milestone generator
- A popular (therefore supported) tool

UML is a language for

- Visualizing
- Specifying
- Constructing and
- Documenting

  the artifacts of a software-intensive systems.

What UML is not!

- A visual programming language or environment
- A database specification tool
- A development process (i.e. an SDLC)

What UML can do for you

- Helps you to:–
- Better think out and document your system before implementing it
- “forecast” your system
- Determine islands of reusability
- Lower development costs

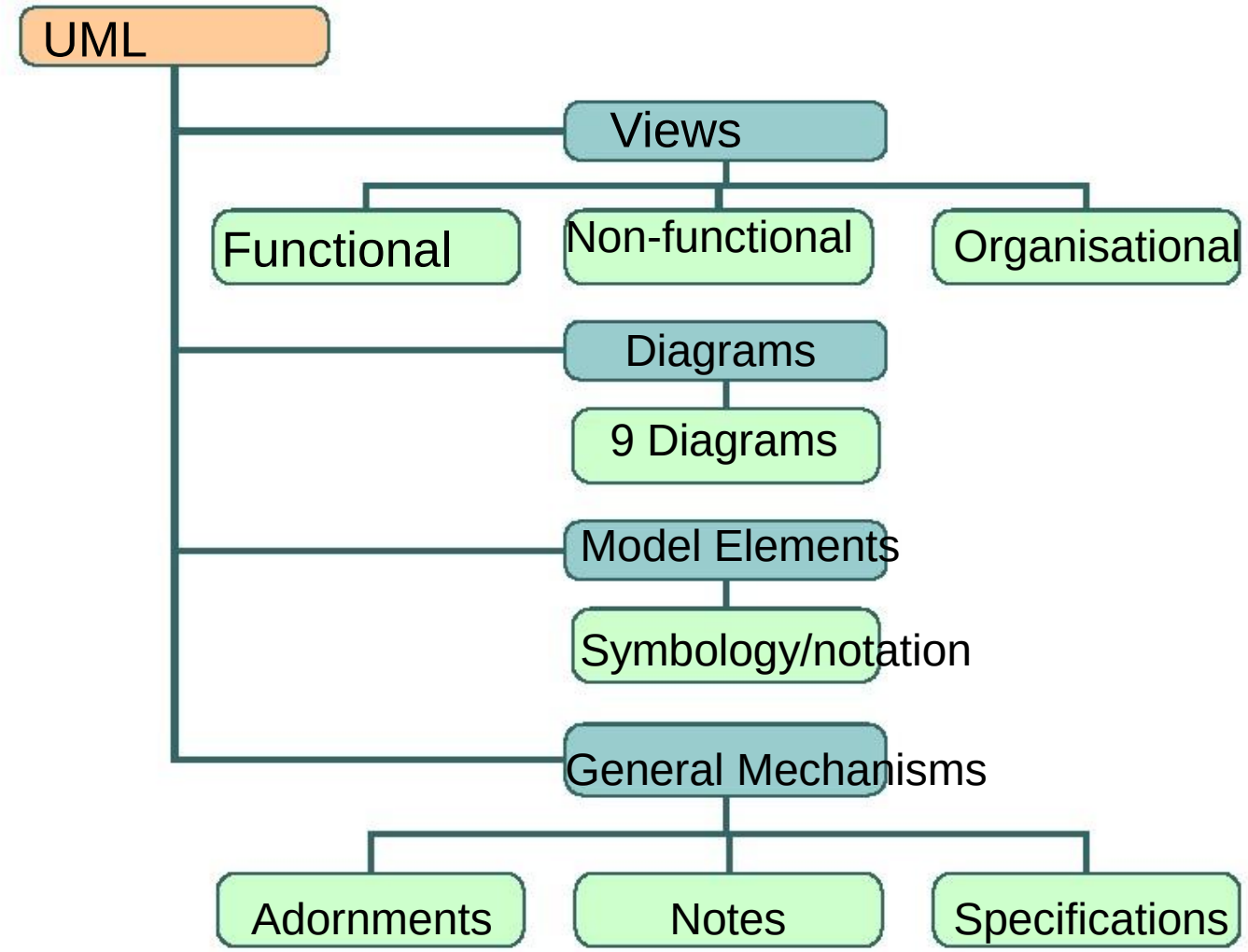
Contd…

- Make the right decisions at an early stage (before committed to code)
- Better deploy the system for efficient memory and processor usage
- Plan and analyse your logic (system behaviour)

UML Partners

- Hewlett-Packard
- IBM
- Microsoft
- Oracle
- i-Logix
- Intelli Corp.
- MCI Systemhouse
- ObjectTime and many more

UML Components



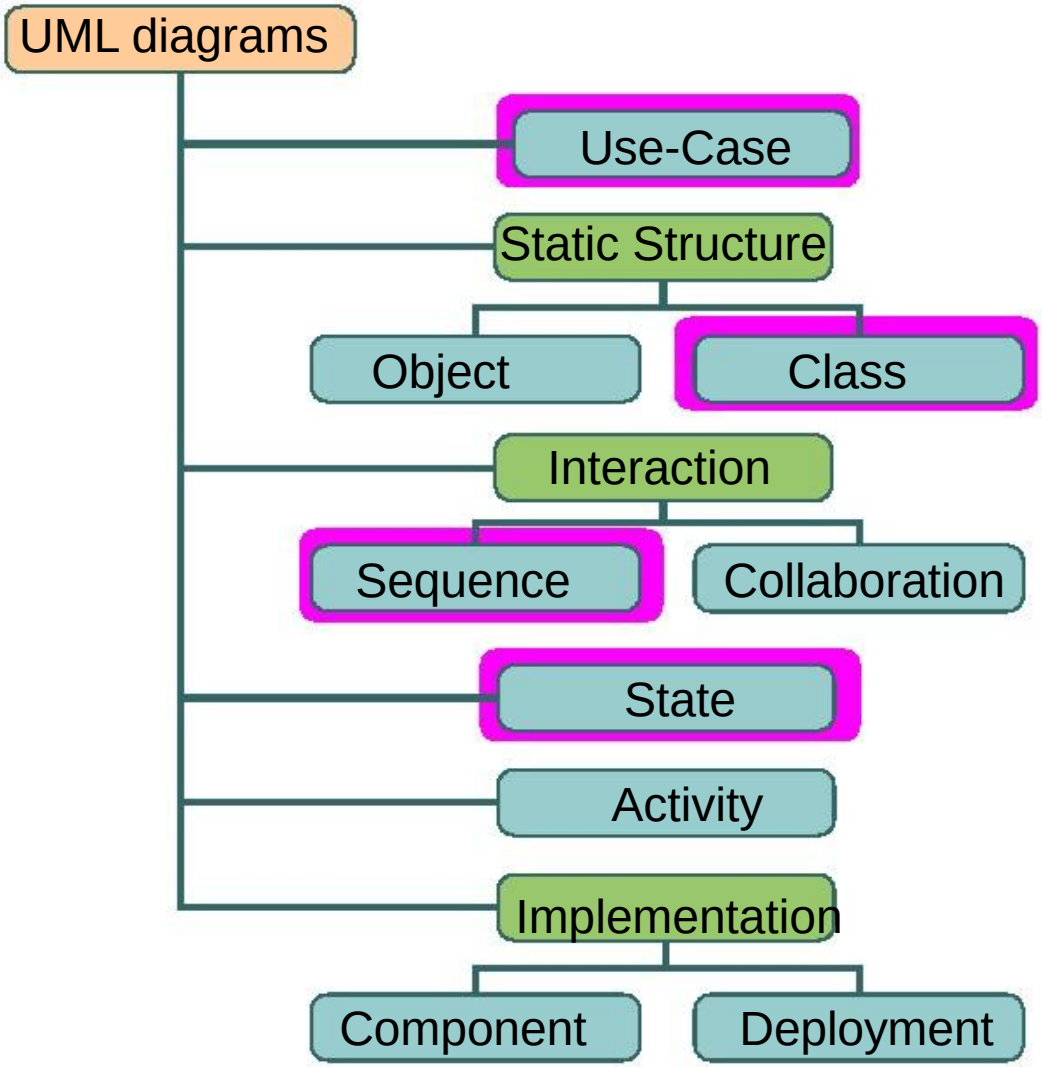
The Case “for” Diagrams

- Descriptive
- Compressive
- Simple
- Understandable
- Universal
- Formalise-able / Standardise-able

The Case “against” Diagrams

- Not inherent knowledge
- Easily cluttered
- Require some training
- Not necessarily revealing
- Must be liked to be accepted and used
- Effort to draw

UML DIAGRAMS CLASSIFICATION



UML Diagrams (comparative slide)

- Use-Case (relation of actors to system functions) E.g.
- Class (static class structure) E.g.
- Object (same as class - only using class instances– i.e. objects) E.g.
- State (states of objects in a particular class) E.g.
- Sequence (Object message passing structure) E.g.

Contd…..

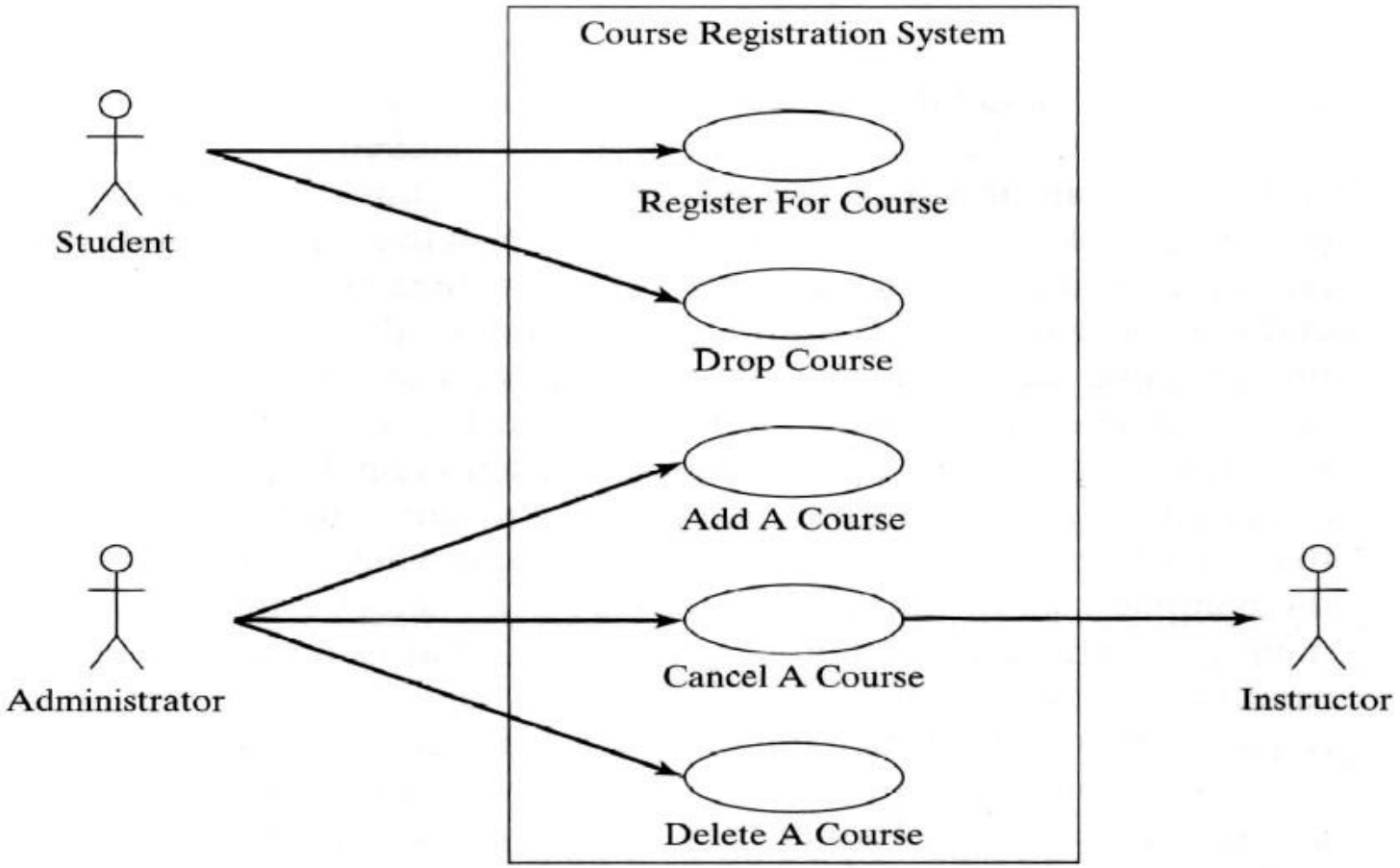
- Collaboration (same as sequence but also shows context - i.e. objects and their relationships) E.g.
- Activity (sequential flow of activities i.e. action states) E.g.
- Component (code structure) E.g.
- Deployment (mapping of software to hardware) E.g.

UML Diagram Philosophy

Any UML diagram:

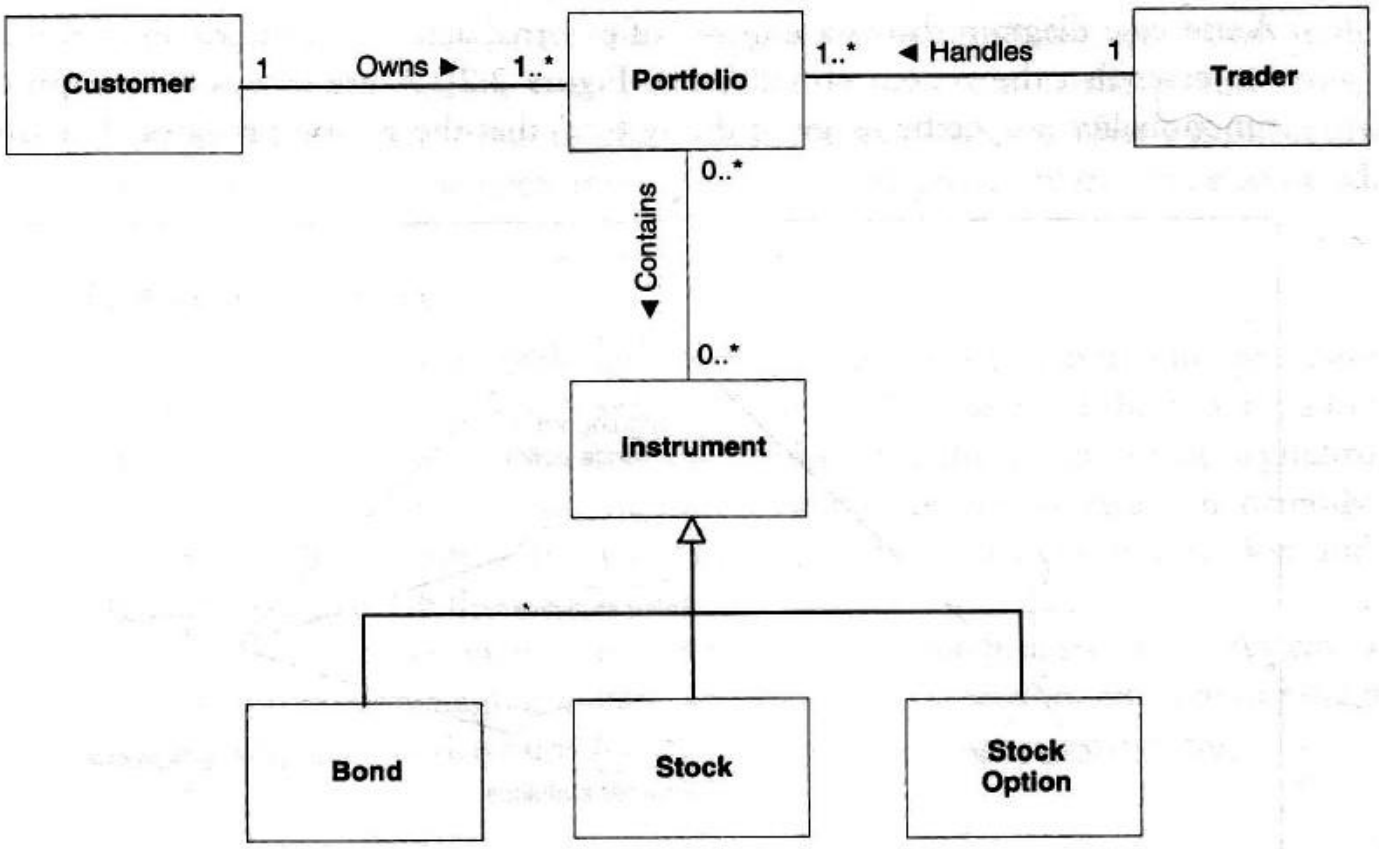
- Depicts concepts– as symbols
- Depicts relationships between concepts– as directed or undirected arcs (lines)
- Depicts names– as labels within or next to symbols and lines

Use Case Diagram



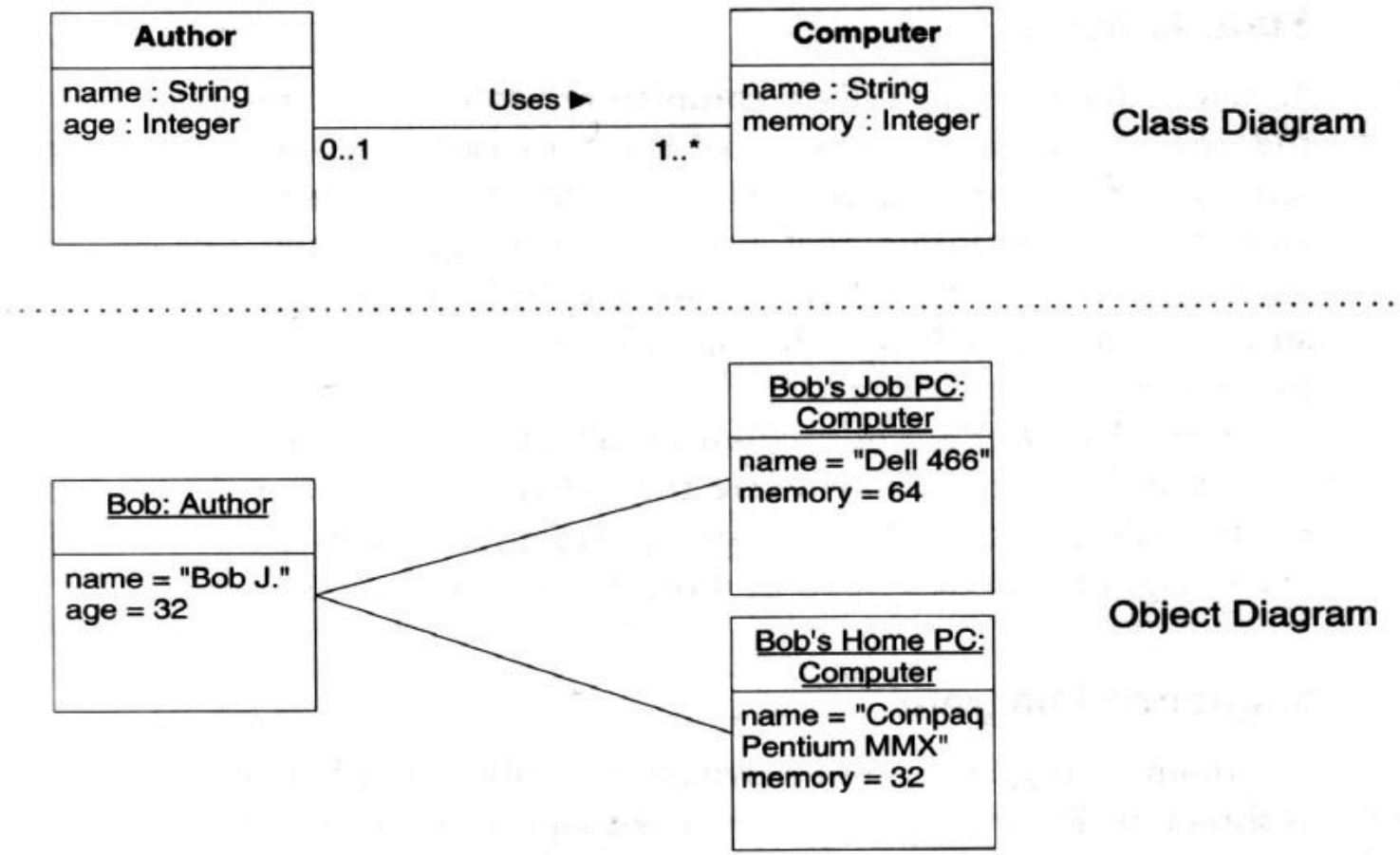
Back

Class Diagram



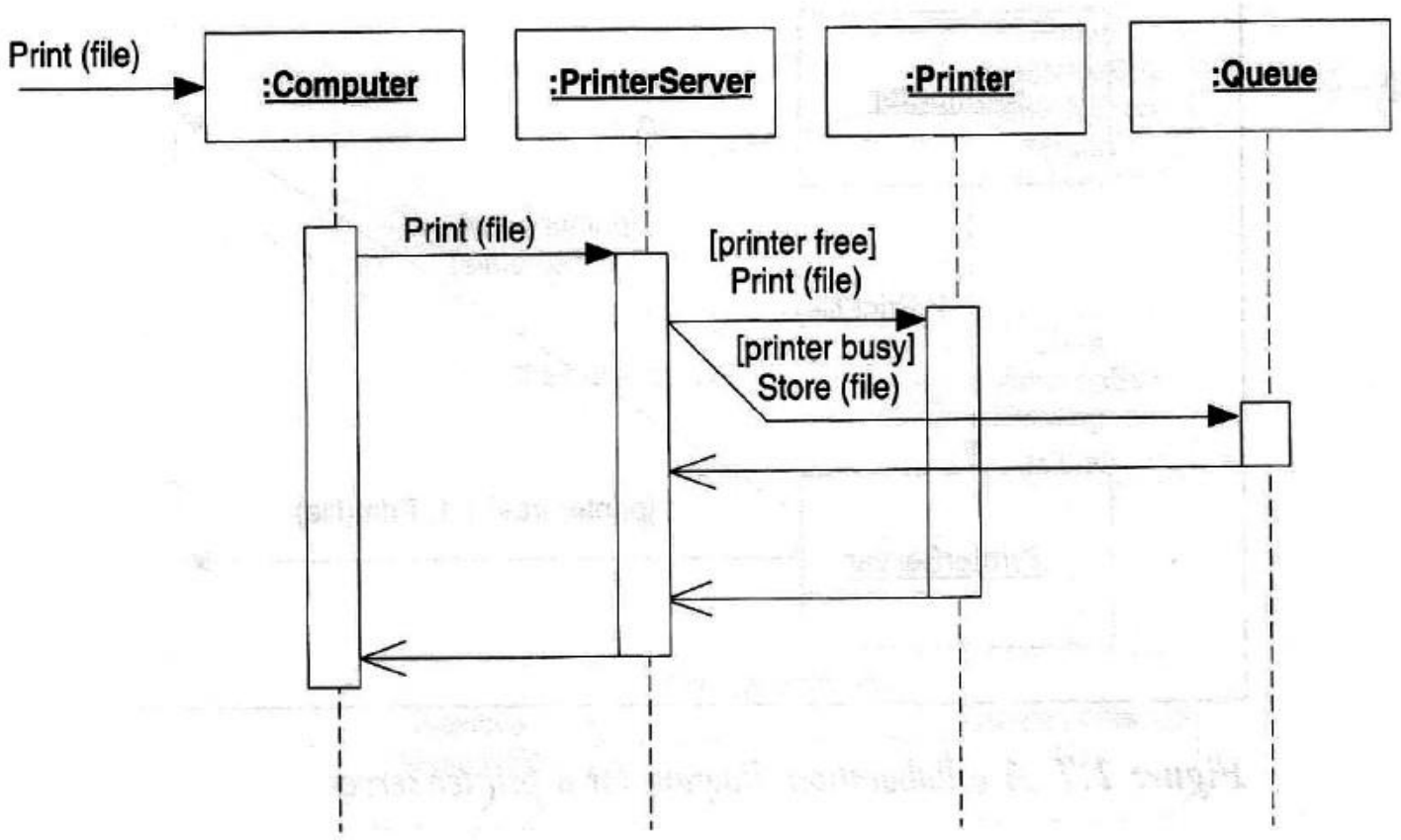
Back

Object Diagram



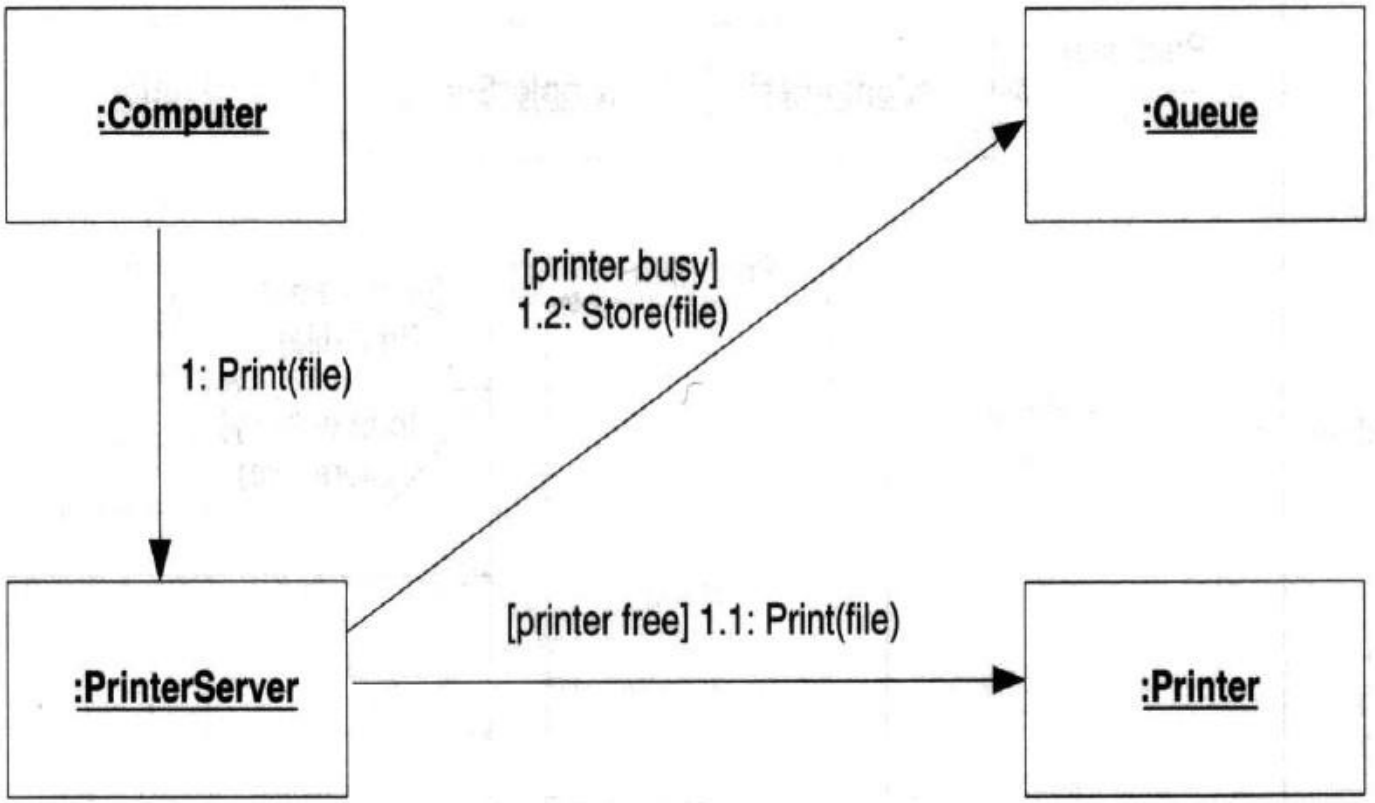
Back

Sequence Diagram



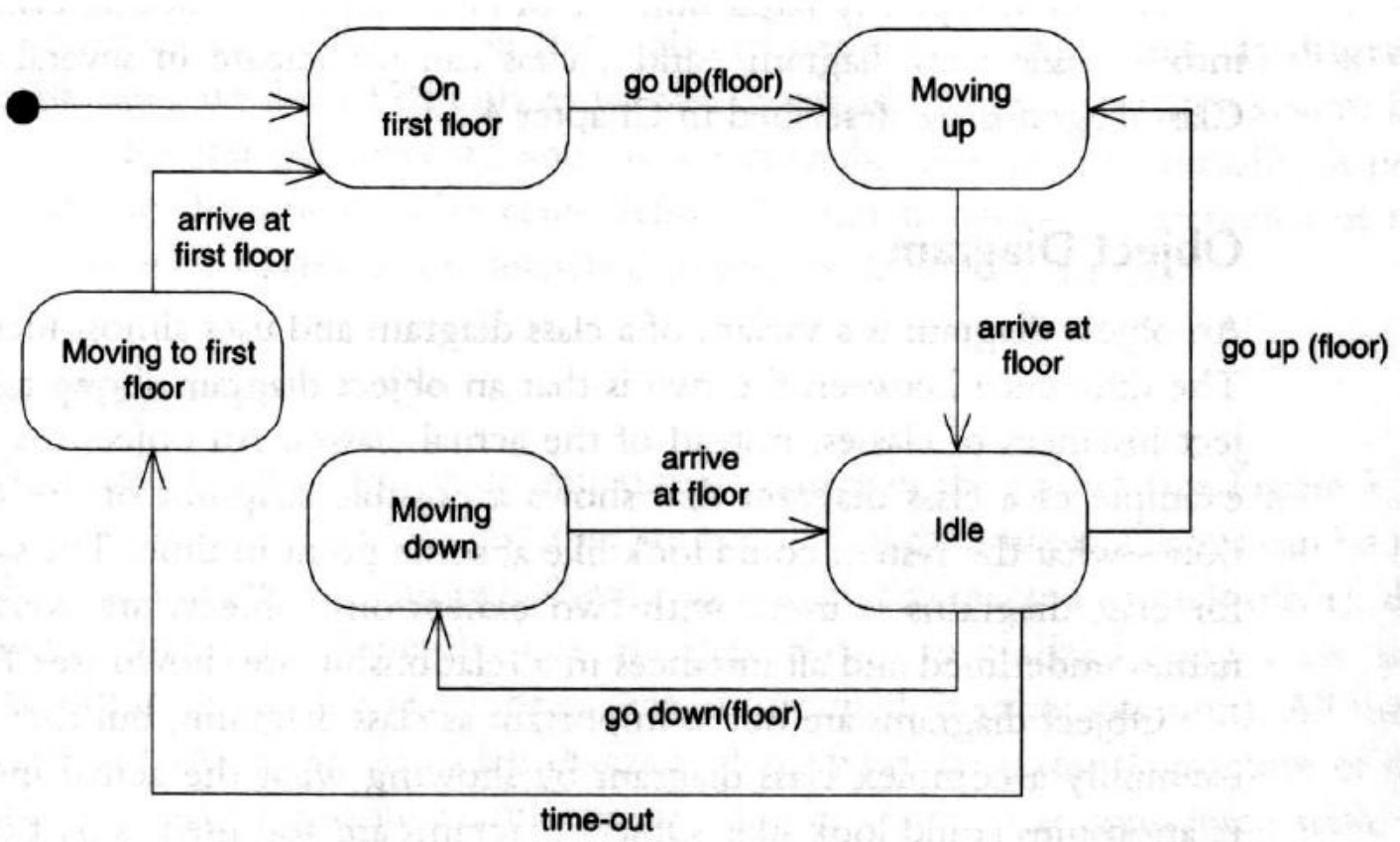
Back

Collaboration Diagram



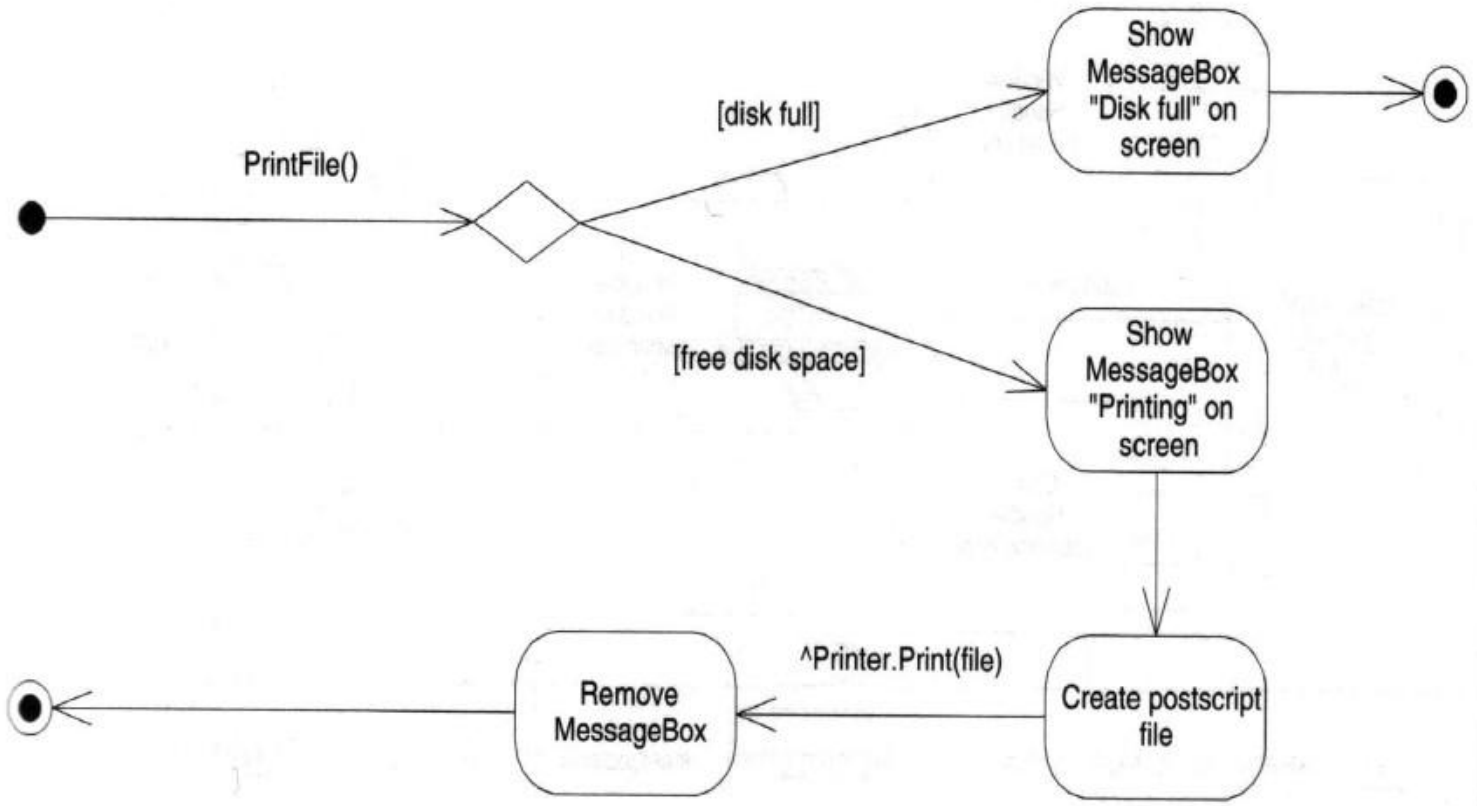
Back

State Diagram



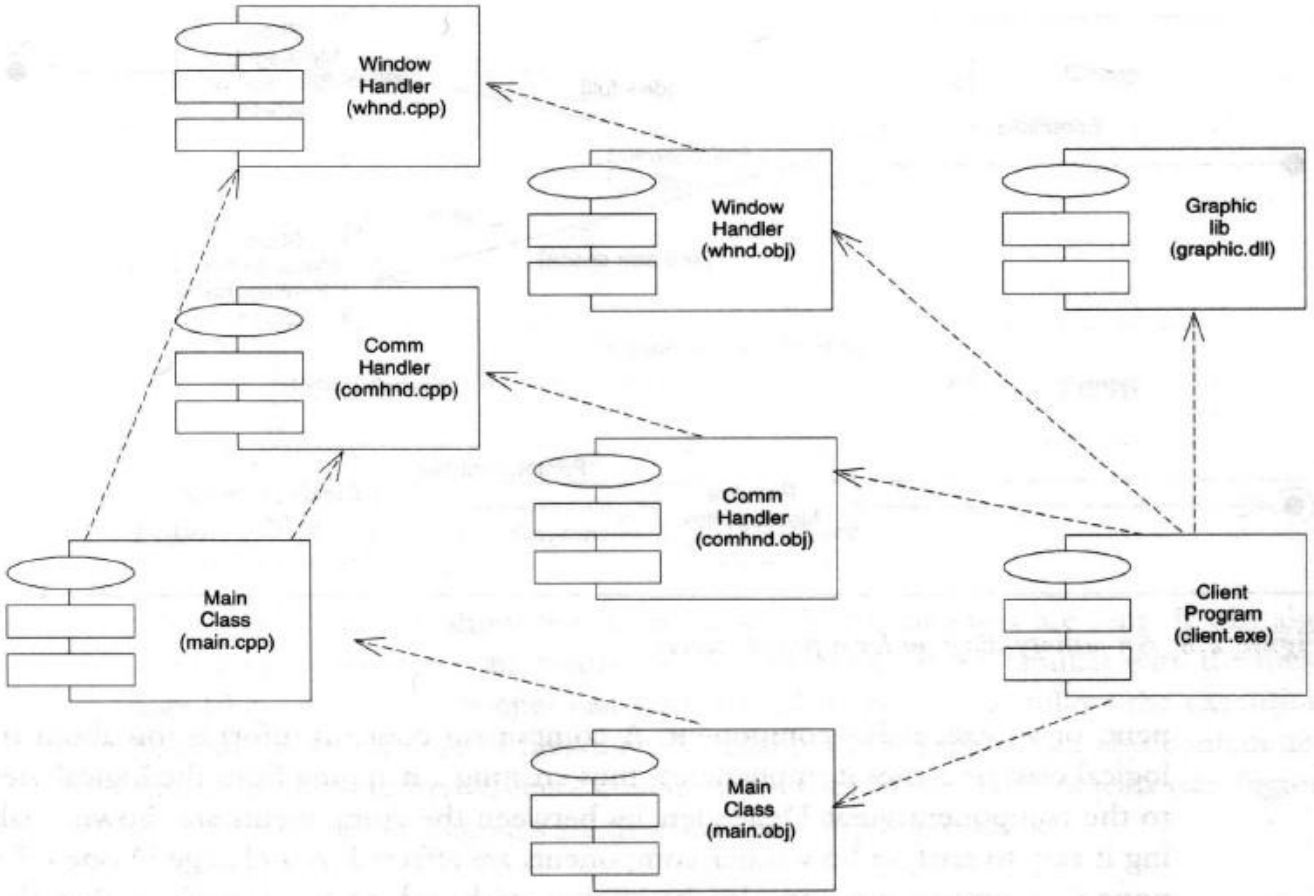
Back

Activity Diagram



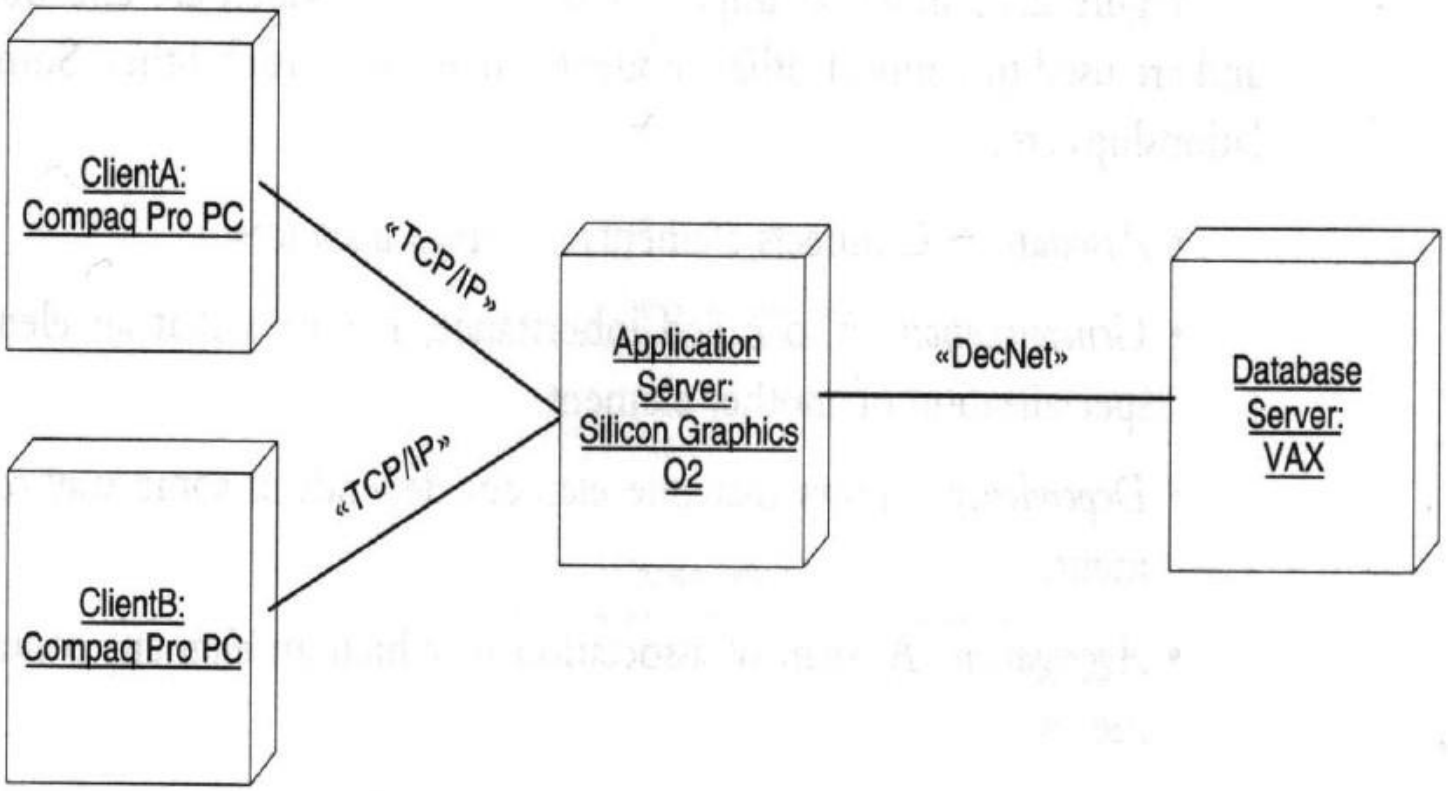
Back

Component Diagram



Back

Deployment Diagram



Back

Review Questions

- Define :
  - OOA
  - OOD
  - OOAD
  - OML
- Define Modeling . What are its objectives?
- What is UML? Explain role of different UML diagrams in system design? [May 2016]
- What is UML? List different UML diagrams.